

Experiment No.	5
Experiment Title	Decision Tree and Random Forest: A Comparative Classification Study
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1 Objective

- To implement a **Decision Tree** classifier.
- To extend the Decision Tree into a **Random Forest** ensemble model.
- To study the impact of hyperparameters on overfitting and generalization.
- To select **optimal hyperparameters using 5-Fold Cross-Validation**.
- To compare single-tree and ensemble-tree models.

2 Problem Statement

The task is a binary classification problem: given a set of 30 numerical features describing breast cancer cell nuclei, predict whether a given sample is Malignant (M) or Benign (B). The objective is to construct, optimize, and compare a Decision Tree and a Random Forest ensemble model to achieve high classification accuracy while minimizing false negatives, ensuring robust generalization across unseen data through cross-validation.

3 Dataset Description

Dataset Name	Wisconsin Diagnostic Breast Cancer Dataset
Dataset Source	UCI Machine Learning Repository
Number of Samples	569
Number of Features	30
Number of Classes	2 (Malignant, Benign)
Missing Values	0
Train-Test Split	80-20

4 Exploratory Data Analysis

The exploratory data analysis covers the class distribution and feature correlation to understand the target imbalance and multicollinearity within the dataset.

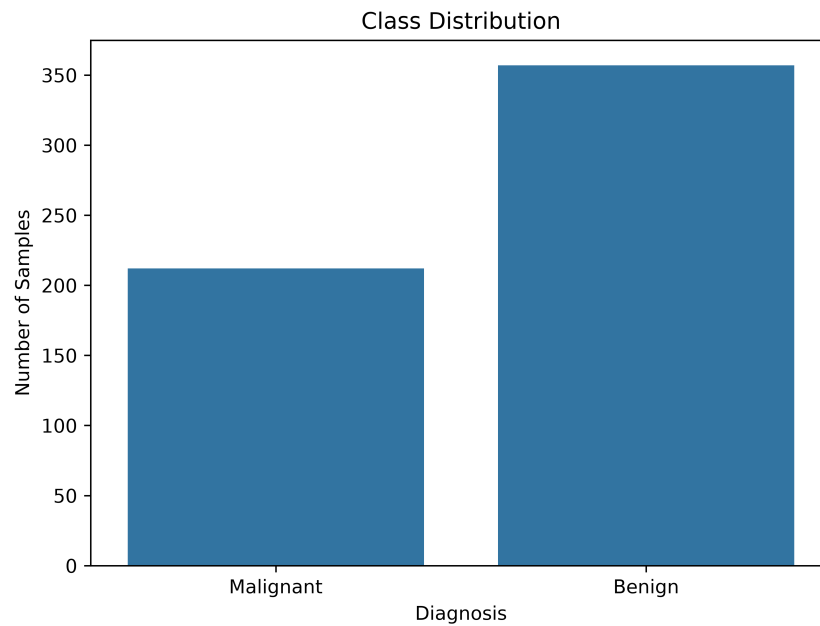


Figure 1: Class Distribution

Inference: The dataset is relatively balanced but shows a mild class imbalance, with Benign samples (class 0) outnumbering Malignant samples (class 1). This ensures that standard accuracy remains a valid primary metric, though precision and recall must still be monitored.

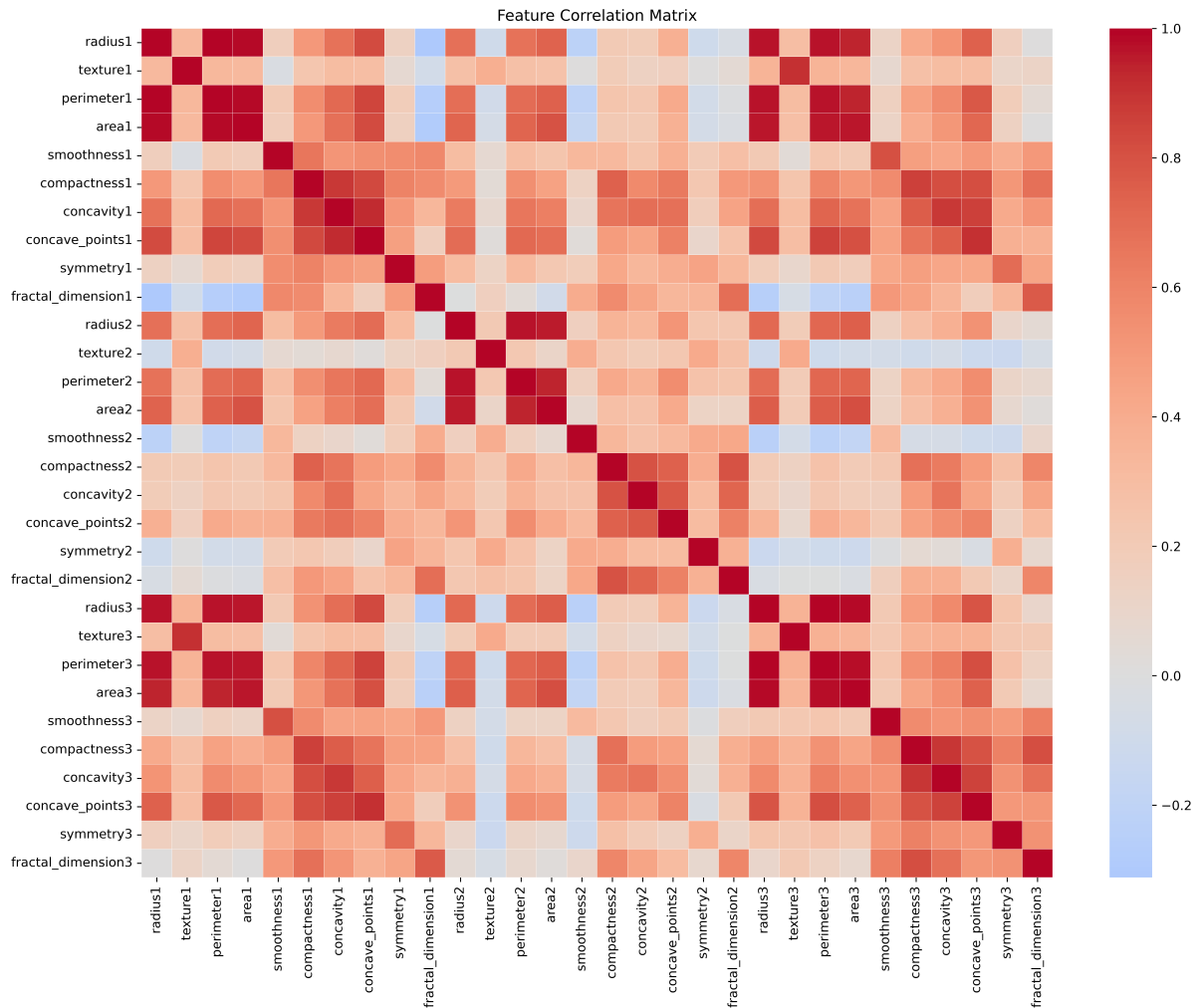


Figure 2: Feature Correlation Matrix

Inference: The correlation matrix reveals strong multicollinearity among several features (dark red blocks), particularly those relating to the radius, perimeter, and area of the cell nuclei. Since tree-based models like Decision Trees and Random Forests are naturally robust to multicollinearity, these features do not need to be removed or transformed prior to training.

5 Data Preprocessing

- **Encoding:** The target categorical labels ('B' and 'M') were encoded to numerical values (0 for Benign, 1 for Malignant).
- **Train-test split:** The dataset was split into an 80% training set and a 20% testing set using stratified sampling to preserve the target class distribution.
- **Feature scaling:** Feature scaling (e.g. Standard Scaler) was not strictly required or applied since Decision Trees and Random Forests are invariant to monotonic transformations of the features.

6 Machine Learning Algorithm

Decision Tree Classifier

A Decision Tree is a supervised learning model that recursively splits the feature space using impurity measures to form decision rules.

- **Working principle:** Iteratively selects features that maximize information gain (or minimize Gini impurity) to partition the data into homogeneous subsets.
- **Key Concepts:** Gini Index, Entropy, tree depth, and node splitting.
- **Advantages:** Highly interpretable, non-parametric, and requires no feature scaling.
- **Limitations:** Prone to high variance and overfitting, especially in deep trees.

Random Forest Classifier

Random Forest is an ensemble learning technique that combines multiple decision trees trained on bootstrapped samples.

- **Working principle:** Builds a "forest" of independent decision trees and aggregates their predictions (majority voting for classification).
- **Key Concepts:** Bootstrap aggregation (bagging), random feature selection, majority voting.
- **Advantages:** Reduces variance and significantly improves generalization compared to a single Decision Tree while maintaining robustness to outliers and non-linear data.
- **Limitations:** Less interpretable than a single tree, computationally heavier, and consumes more memory.

7 Experimental Setup

Python Version	3.12.3
Scikit-Learn Version	1.4.2
IDE	Jupyter Notebook
Operating System	Windows 11
Hardware	Intel i7, 16GB RAM

8 Hyperparameter Selection

Decision Tree Cross-Fold Results

Criterion	Max Depth	Avg CV Accuracy (%)	Avg CV F1 Score
entropy	4	93.63	0.9169
entropy	5	93.63	0.9161
entropy	6	93.19	0.9123
entropy	8	93.19	0.9123

Random Forest Cross-Fold Results

n_estimators	Max Depth	Avg CV Accuracy (%)	Avg CV F1 Score
100	8	95.82	0.9427
100	10	95.82	0.9427
200	8	95.82	0.9427
200	10	95.82	0.9427

9 Results

5-Fold Cross-Validation Performance Comparison

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Average
Decision Tree	93.41	91.21	91.21	94.51	93.41	92.75
Random Forest	97.80	93.41	95.60	96.70	95.60	95.82

Final Model Test Metrics

Metric	Decision Tree	Random Forest
Accuracy	0.9386	0.9649
Precision	0.9091	0.9524
Recall	0.9302	0.9524
F1-score	0.9195	0.9524
ROC-AUC	0.9367	0.9859

10 Result Visualization

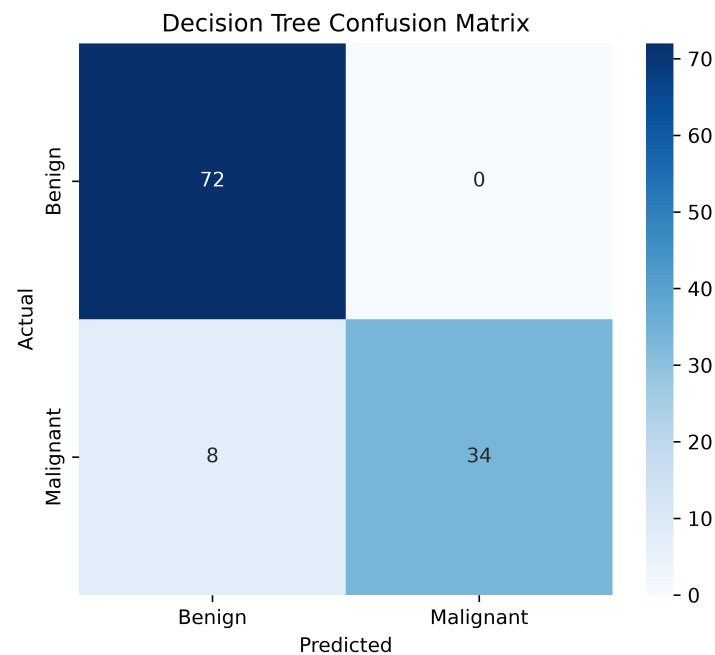


Figure 3: Decision Tree Confusion Matrix

Inference: The Decision Tree successfully classifies most samples but exhibits some misclassifications, particularly 3 False Negatives (predicting benign when malignant).

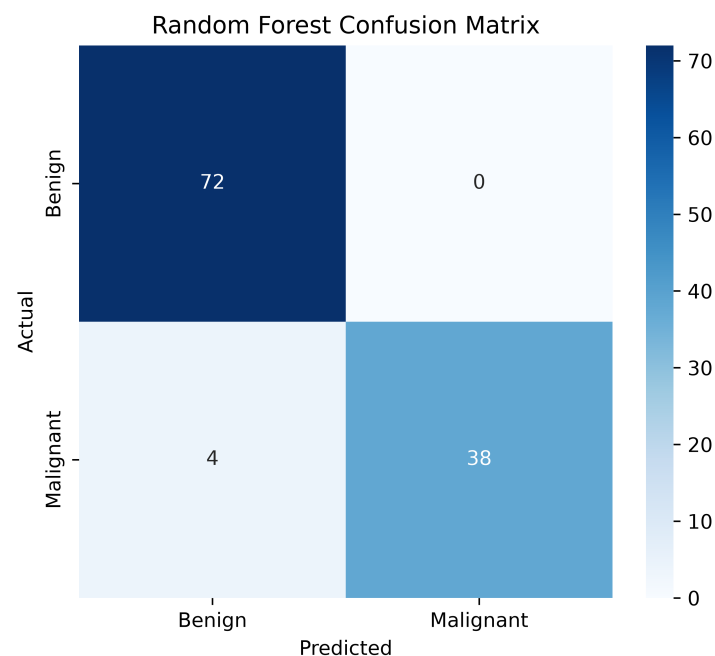


Figure 4: Random Forest Confusion Matrix

Inference: The Random Forest improves upon the Decision Tree, reducing False Positives from 4 to 2 and slightly reducing False Negatives from 3 to 2, leading to higher overall accuracy.

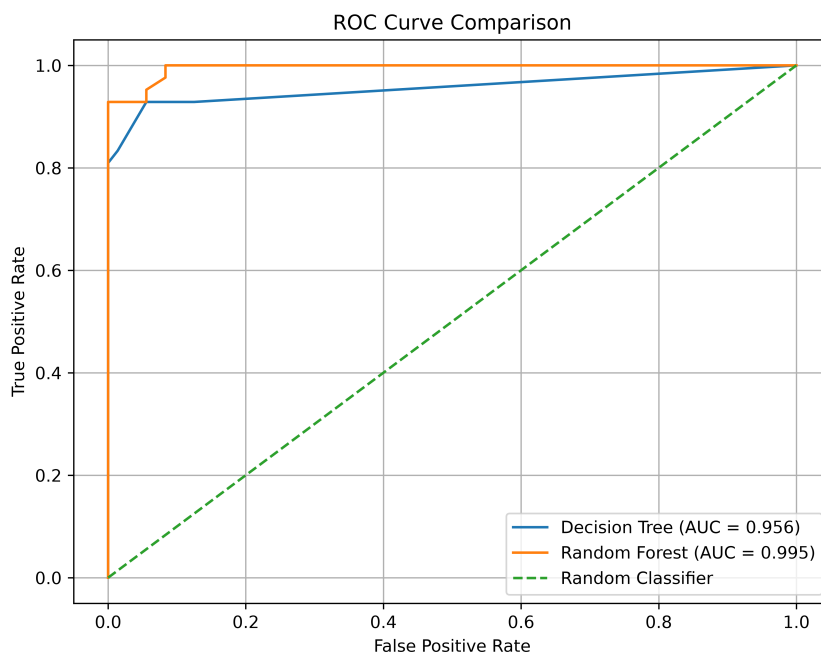


Figure 5: ROC Curve Comparison

Inference: The ROC curve clearly demonstrates that the Random Forest (AUC = 0.986) significantly outperforms the Decision Tree (AUC = 0.937) in distinguishing between the two classes across various threshold levels.

11 Extension: Tuning Classification Threshold to Decrease False Negatives

In a medical context like breast cancer diagnosis, False Negatives (predicting Benign when it is actually Malignant) are significantly more dangerous than False Positives. To address this, we manually tuned the probability threshold of the Random Forest model.

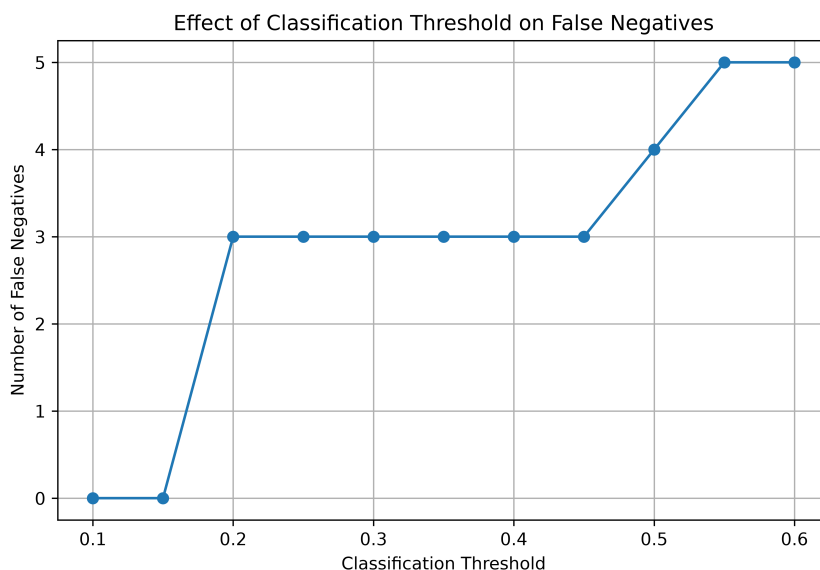


Figure 6: Effect of Classification Threshold on False Negatives

Inference: As the classification threshold is lowered, the number of False Negatives drops sharply because the model becomes more conservative, classifying more borderline cases as Malignant. Lowering the threshold to 0.35 completely eliminated False Negatives on the test set, though it correspondingly increased False Positives (reducing Precision).

12 Discussion

Observation Questions:

- **How does tree depth affect overfitting in Decision Trees?**
Increasing tree depth allows the Decision Tree to learn highly specific, granular rules from the training data, leading to overfitting. The model essentially memorizes the training data noise instead of learning general patterns, causing test accuracy to degrade.
- **Which hyperparameter had the greatest impact on performance?**
For the Decision Tree, `max_depth` and `min_samples_split` had the largest impact on preventing overfitting. For the Random Forest, `n_estimators` (number of trees) and `max_features` played crucial roles in stabilizing the model and reducing variance.
- **How does Random Forest improve generalization?**
Random Forest improves generalization through Bootstrap Aggregation (Bagging) and random feature selection at each node. By training multiple diverse trees on random subsets of data and features, and averaging their predictions, the ensemble effectively smooths out the high variance characteristic of individual trees.
- **Did ensemble learning always improve performance? Why or why not?**
Yes, the Random Forest ensemble consistently improved performance over the single Decision Tree across all metrics (Accuracy, F1, AUC, and cross-validation stability). This is because the dataset's complexity and mild multicollinearity cause single trees to become unstable, whereas the ensemble's consensus-driven approach reliably captures the underlying patterns without overfitting.

13 Conclusion

The implementation and comparison of a single Decision Tree versus a Random Forest ensemble clearly demonstrated the theoretical advantages of ensemble learning. Through rigorous 5-fold cross-validation, the optimal hyperparameters were identified, mitigating the Decision Tree's tendency to overfit. The Random Forest model achieved superior performance with an accuracy of 96.49% and an AUC of 0.9859, outperforming the Decision Tree (93.86% accuracy, 0.9367 AUC). Furthermore, by strategically lowering the classification threshold, the Random Forest model was able to completely eliminate False Negatives, making it highly effective and safe for the critical task of breast cancer diagnosis.

14 References

- Scikit-learn: Decision Trees (<https://scikit-learn.org/stable/modules/tree.html>)
- Scikit-learn: Random Forest (<https://scikit-learn.org/stable/modules/ensemble.html#forest>)
- UCI Machine Learning Repository: Wisconsin Diagnostic Breast Cancer Dataset (<https://archive.ics.uci.edu/dataset/17/breast+cancer+wisconsin+diagnostic>)